NIET Greater Noida	**Data Structures And Algorithm-II (BCSE-0401) UNIT-5**	**SESSION: 2025-26** **COURSE/SEM: B.TECH/IVth**

Assignment Given Date: 18/04/2026 Assignment Submission Date: 25/04/2026	**MAXIMUM MARKS: 10**
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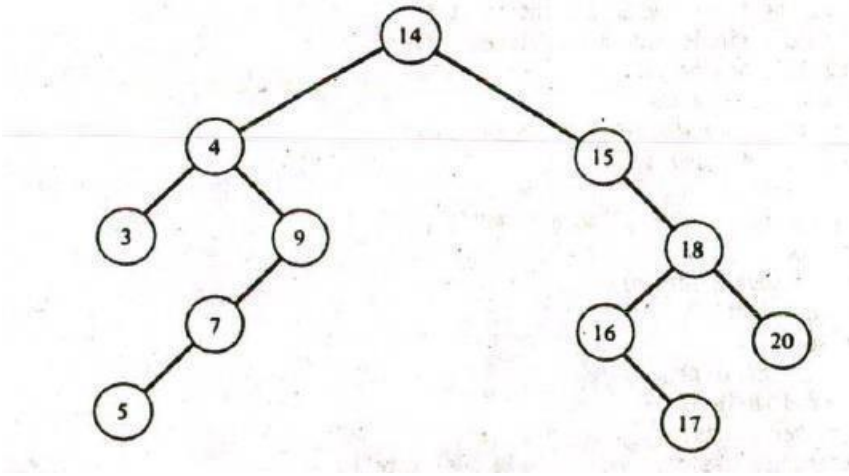
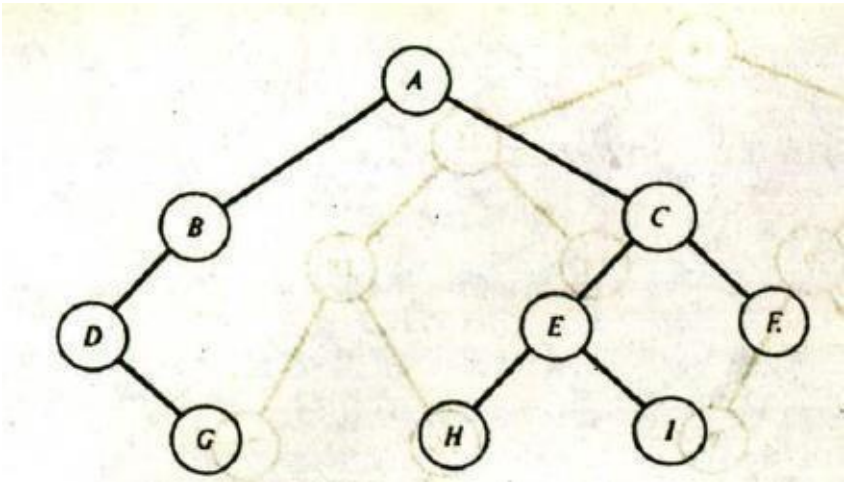
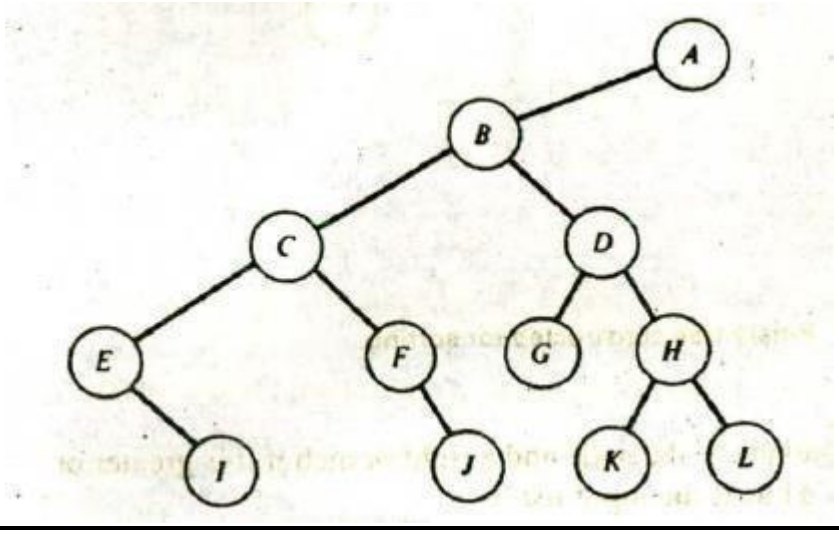
INSTRUCTIONS:

1. Submit the assignment before the due date.
2. Avoid copying the assignment, if found student will be penalized accordingly.
3. Mention all the necessary details in your pdf file. All assignments should be in a pdf file format. No other format will be entertained.

Q. No.	Question Statement	CO	BLOOM'S KNOWLEDGE LEVEL
1.	**Q1. Red-Black Trees** Given the sequence of keys: `20, 15, 25, 10, 5, 1` • Insert the keys one by one into a Red-Black Tree. • Show all rotations and recoloring steps. • Draw the tree after each insertion. • Verify all Red-Black Tree properties at the end.	**CO5**	**K4**
2.	**Q2. B-Tree** Consider a B-Tree of order **4**. • Insert the following keys: `10, 20, 5, 6, 12, 30, 7, 17` • Show step-by-step insertion. • Draw the B-Tree after each insertion. • Explain node splitting clearly.	**CO5**	**K3**
3.	**Q3. B+ Tree** Construct a B+ Tree of order **3** for the keys: `5, 15, 25, 35, 45, 55, 65` **i.** Insert all keys step by step. **ii.** Show how leaf nodes are linked. **iii.** Compare the final structure with a B-Tree built using the same keys. **iv.**	**CO5**	**K3**

4.	**Q4. Binomial Heap** Perform the following operations on an empty Binomial Heap: • Insert keys: `10, 20, 30, 5, 7` • Perform **Extract-Min** • Show the structure of the heap after each operation. • Explain how binomial trees are merged. •	**CO5**	**K2**
5.	**Q5. Fibonacci Heap** For a Fibonacci Heap, perform the operations: **i.** Insert keys: `7, 3, 17, 24, 10` **ii.** Perform **Extract-Min** **iii.** Perform **Decrease-Key (17 → 2)** **iv.** Show the heap structure after each step. **v.** Explain cascading cuts and why Fibonacci Heap is efficient.	**CO5**	**K3**

6.	**Find the In-Order, Pre-Order and Post-Order Traversal for the following Binary Tree:**    	**CO1**	**K2**

7.	**Create a Binary Tree from the given traversal:** i. Pre-Order: 1 2 4 5 3 6 In-Order: 4 2 5 1 6 3 ii. In-Order: 40 20 50 10 60 30 Post-Order: 40 50 20 60 30 10 iii. Pre-Order: A B D I J F C G K H In-Order: I D J B F A G K C H	CO1	K2
8.	**Define Binary Search Tree. Give a program to implement BST and to insert and delete in BST.**	CO1	K3
9.	**Construct a Binary Search Tree by inserting the following sequence of number: 10,12,5,4,20,8,7,15 and 13.** **After creating the binary tree delete the following node: 13, 5 and 10 from the above binary search tree created.**	CO1	K2
10.	**Create an AVL Tree from the given nodes:** i. 21,26,30,9,4,14,28,18,15,10,2,3,7 ii. Create an AVL tree by inserting numbers from 1 to 10. iii. Create an AVL tree by inserting the following sentence: ***"Jump dogs bad".***	CO1	K4
11.	**Perform Deletion in AVL Tree from the tree created above in the sequence mentioned below:** i. 2, 15, 9, 21,14 ii. 8, 7, 2, 6	CO1	K3
12.	**Discuss the following with proper example:** i. Binary Heaps ii. Priority Queue	CO1	K2
13.	**Perform Heap Sort on the following listed numbers and give the algorithm to perform heap sort for the same.** **82, 90, 10, 12, 15, 77, 55, 23**	CO1	K2
14.	**What are Threaded Binary Trees. Discuss its advantages and disadvantages.**	CO1	K2